

An Unofficial Compendium of Worked Solutions to the Problems in *Geometry*

by I. M. Gelfand & T. Alekseyevskaya (Gelfand)

Deepak Venkatesh

September 8, 2026

Abstract

This book is an independent and unofficial collection of worked solutions to problems inspired by the book *Geometry* by I. M. Gelfand and T. Alekseyevskaya (Gelfand), first published in 2020.

The original text presents a wide range of problems in projective geometry, affine geometry, symplectic geometry, and Euclidean geometry. The book contains solved exercises and questions, but the reader is also given 230 problems to solve.

The aim of this book is to provide clear, logically consistent solutions to all of the 230 unsolved problems addressed, written in a style accessible to motivated students and self-learners.

Figures are either drawn by hand and scanned, constructed using GeoGebra, or produced directly in $\text{\LaTeX}$. Where applicable, proofs or calculations may also be checked using computer programs written in Scheme (a dialect of Lisp) or Python.

Large language models were used only as auxiliary tools, primarily for generating figures in $\text{\LaTeX}$ and for reviewing exposition.

This work is not affiliated with, endorsed by, or approved by the original authors or publisher. Any errors, omissions, or inconsistencies are entirely my own.

Issues and corrections may be reported at the [project repository](#).

Copyright and Disclaimer

Unofficial companion.

This document is an **independent, unofficial collection of worked solutions** to problems from *Geometry* by I.M. Gelfand and T. Alekseyevskaya (Gelfand) (first published in 2020).

This work is **not affiliated with, endorsed by, or approved by** the original authors or the publisher. The names of the authors and the title of the book are used only for identification and reference.

The original problem statements are not reproduced here. All exposition and solutions are written independently by the author of this document.

Copyright © 2026 Deepak Venkatesh

License:

This work is licensed under a [Creative Commons “Attribution-NonCommercial-ShareAlike 4.0 International”](#) license.



Project repository:

<https://github.com/deepak-venkatesh/gelfand-geometry>

For
My Daughter and Son

Contents

Working Notes	7
Introduction	8
 Part I: Points and Lines: A Look at Projective Geometry	 9
Chapter 1: Points and lines	10
Chapter 2: Two lines and an angle	14
Chapter 3: Three lines	15
Chapter 4: Four lines. Quadrilaterals	16
Chapter 5: Five lines	17
Chapter 6: Projection from a point onto a line	18
Chapter 7: Dual configurations in projective geometry	19
Chapter 8: Desargues configuration	20
Chapter 9: Dual Desargues configuration	21
Chapter 10: Algebraic notation of “computer presentation” of configurations	22
Chapter 11: Polygons and n straight lines	23
Chapter 12: Convex polygons, convex hull of n points	24
Chapter 13: Solution of Exercise 3 with the help of a Desargues configuration	25
Chapter 14: Overview of Chapter I	26
 Part II: Parallel Lines: A Look at Affine Geometry	 27
 Part I: Lines and Segments	 28
Chapter 1: Parallel Straight Lines	29
Chapter 2: Operations available in Chapter II	30
Chapter 3: Properties of parallel lines	31
Chapter 4: Segments lying on parallel lines	32
 Part II: Figures	 33

Chapter 5: Parallelograms	34
Chapter 6: Triangles	35
Chapter 7: Trapezoids	36
Part III: Operations with Figures	37
Chapter 8: The Minkowsky addition of two figures	38
Chapter 9: Parallel projection	39
Chapter 10: Parallel translation	40
Chapter 11: Central symmetry on the plane	41
Chapter 12: Vectors	42
Chapter 13: Overview of Chapter II	43
Appendix for Chapter II	44
1. Why we cannot define equal segments in Chapter II	44
2. Parallel lines, equal segments, and the Desargues configuration	44
3. Arithmetic operations with segments	44
4. Segments and rational numbers	44
5. Affine coordinate systems on the plane	44
Part III: Area: A Look at Symplectic Geometry	45
Chapter 1: The area of a figure	46
Chapter 2: Area of a parallelogram	47
Chapter 3: Area of a triangle	48
Chapter 4: Area of a trapezoid	49
Chapter 5: Area of a polygon	50
Chapter 6: More problems on areas	51
Chapter 7: How to measure the area of a figure	52
Chapter 8: Overview of Chapter III	53
Part IV: Circles: A Look at Euclidean Geometry	54
Part I: Introduction to the Circle	55
Chapter 1: Operations available in Chapter IV	56

Chapter 2: Comparing segments	57
Chapter 3: Angles	58
Chapter 4: Operations with figures	59
Part II: The Geometry of the Triangle and Other Figures	60
Chapter 5: Elements of triangle. Congruent triangles	61
Chapter 6: Construction of a triangle from its elements	62
Chapter 7: Relations between elements of a triangle	63
Chapter 8: Properties of a triangle. Particular kinds of triangles	64
Chapter 9: Area in Euclidean geometry	65
Chapter 10: The Pythagorean theorem and its applications	66
Chapter 11: Relations between lines and points	67
Chapter 12: Special lines and special points in a triangle	68
Chapter 13: Polygons	69
Chapter 14: Summary of facts about different quadrilaterals	70
Chapter 15: Similarity	71
Part III: Circles	72
Chapter 16: Circles and points	73
Chapter 17: Circles and lines	74
Chapter 18: Two or more circles	75
Chapter 19: Circles and angles	76
Chapter 20: A circle and a triangle	77
Chapter 21: Circles and polygons	78
Chapter 22: Circumference and arc	79
Chapter 23: Disks and sectors	80
Chapter 24: Overview of Chapter IV	81

Working Notes

Challenging Problems for Middle Schoolers:

Typographical Errors:

1. The standard mathematical spelling of Hermann Minkowski's name is *Minkowski* and not *Minkowsky*.

Introduction

No problems in this chapter.

Part I

Points and Lines: A Look at Projective Geometry

Chapter 1. Points and lines

Problem 1.

In a line if we place a point we divide it into 2 parts. In this case both the part are *rays* and the point is an *endpoint*.

Now if there are 2 point on the line, we still have 2 rays but the two endpoints create a line segment.

So n points on a line will give 2 rays and $(n - 1)$ line segments.

For this problem.

$$\begin{aligned} n &= 100 \\ \text{endpoints} &= 100 \\ \text{rays} &= 2 \\ \text{segments} &= 100 - 1 = 99 \end{aligned}$$

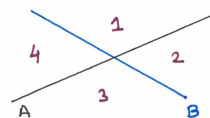
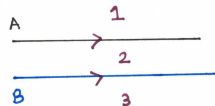
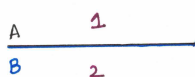


Problem 2.

Three possibilities exist:

1. The two lines A and B can perfectly coincide.
2. They can be parallel to each other.
3. They can intersect at some point.

Basis the figure below the plane can be divided into 2, 3, or 4 parts.



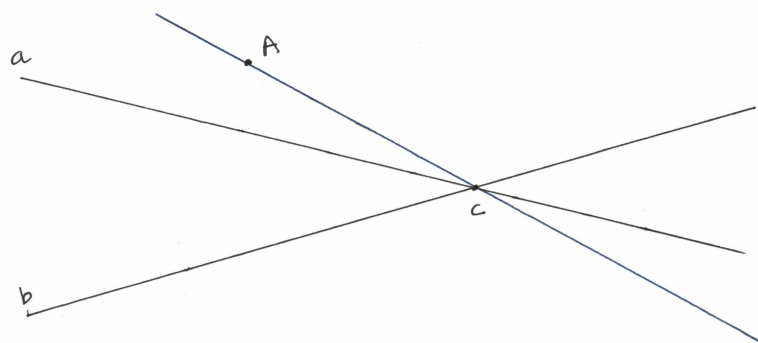
Problem 3.

The chapter explains that geometric constructions performed using the two axioms below are call constructions in *projective geometry*.

Axiom 1: Through any two points, one can draw a unique line.

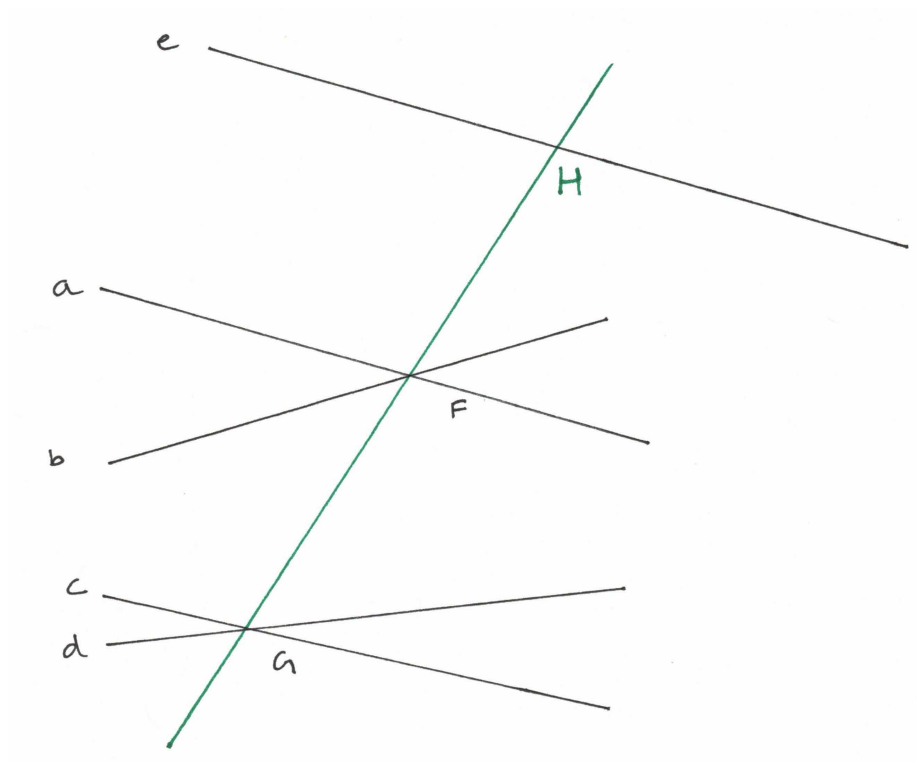
Axiom 2: Any two lines either have a unique point of intersection or do not intersect (are parallel).

Basis these two axioms of projective geometry we can construct the line AC .



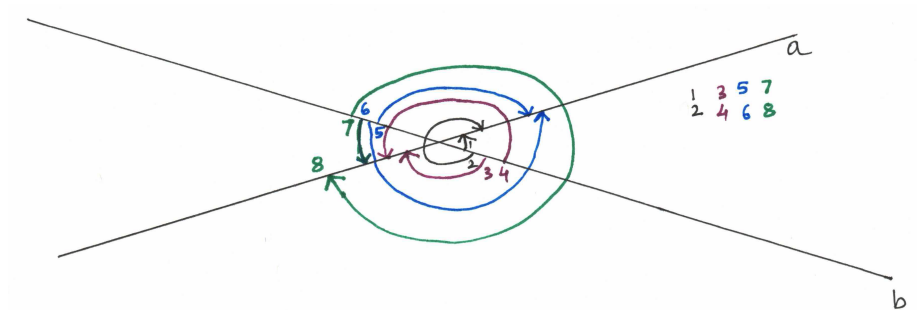
Problem 4.

F and G can be connected with a line since basis Axiom 1 of Projective Geometry we can construct a line. This line intersects at a unique point H with the line e . If e was parallel to line GF then they would never intersect. But given the visual in the problem we are assuming that they will intersect.



Problem 5.

There are 8 angles in this. From one side of the ray of b to one side of a (via convex) and another via not the convex side. Similarly 2 ways to other side ray a . Then 4 more ways for the other side of ray b .



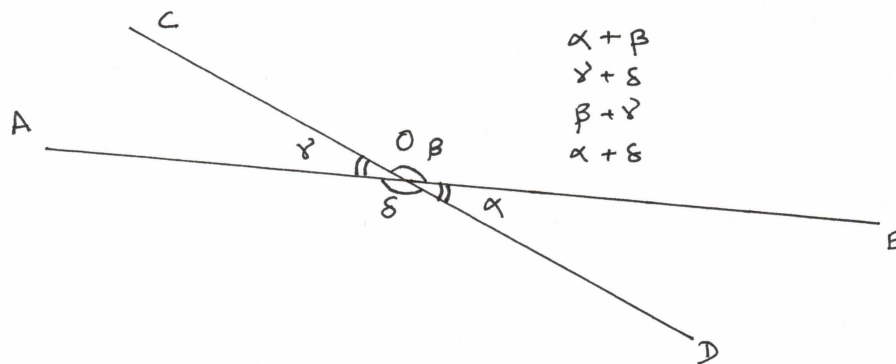
Problem 6.

The definition of supplementary angles is important. To quote the authors

Two convex angles that have one side in common are positioned in such a way that the second side of one angle is an extension of the second side of the other angle are called *supplementary angles*.

In schools we are usually taught that if the sum of two angles is 180° they are called supplementary angles.

1. $\angle AOC$ and $\angle COB$
2. $\angle COB$ and $\angle BOD$
3. $\angle BOD$ and $\angle DOA$
4. $\angle DOA$ and $\angle AOC$



Chapter 2. Two lines and an angle

Chapter 3. Three lines

Chapter 4. Four lines. Quadrilaterals

Chapter 5. Five lines

Chapter 6. Projection from a point onto a line

Chapter 7. Dual configurations in projective geometry

Chapter 8. Desargues configuration

Chapter 9. Dual Desargues configuration

Chapter 10. Algebraic notation of “computer presentation” of configurations

Chapter 11. Polygons and n straight lines

Chapter 12. Convex polygons, convex hull of n points

Chapter 13. Solution of Exercise 3 with the help of a Desargues configuration

Chapter 14. Overview of Chapter I

Part II

Parallel Lines: A Look at Affine Geometry

Part I

Lines and Segments

Chapter 1. Parallel Straight Lines

Chapter 2. Operations available in Chapter II

Chapter 3. Properties of parallel lines

Chapter 4. Segments lying on parallel lines

Part II

Figures

Chapter 5. Parallelograms

Chapter 6. Triangles

Chapter 7. Trapezoids

Part III

Operations with Figures

Chapter 8. The Minkowsky addition of two figures

Chapter 9. Parallel projection

Chapter 10. Parallel translation

Chapter 11. Central symmetry on the plane

Chapter 12. Vectors

Chapter 13. Overview of Chapter II

Appendix for Chapter II

1. Why we cannot define equal segments in Chapter II
2. Parallel lines, equal segments, and the Desargues configuration
3. Arithmetic operations with segments
4. Segments and rational numbers
5. Affine coordinate systems on the plane

Part III

Area: A Look at Symplectic Geometry

Chapter 1. The area of a figure

Chapter 2. Area of a parallelogram

Chapter 3. Area of a triangle

Chapter 4. Area of a trapezoid

Chapter 5. Area of a polygon

Chapter 6. More problems on areas

Chapter 7. How to measure the area of a figure

Chapter 8. Overview of Chapter III

Part IV

Circles: A Look at Euclidean Geometry

Part I

Introduction to the Circle

Chapter 1. Operations available in Chapter IV

Chapter 2. Comparing segments

Chapter 3. Angles

Chapter 4. Operations with figures

Part II

**The Geometry of the Triangle and
Other Figures**

Chapter 5. Elements of triangle. Congruent triangles

Chapter 6. Construction of a triangle from its elements

Chapter 7. Relations between elements of a triangle

Chapter 8. Properties of a triangle. Particular kinds of triangles

Chapter 9. Area in Euclidean geometry

Chapter 10. The Pythagorean theorem and its applications

Chapter 11. Relations between lines and points

Chapter 12. Special lines and special points in a triangle

Chapter 13. Polygons

Chapter 14. Summary of facts about different quadrilaterals

Chapter 15. Similarity

Part III

Circles

Chapter 16. Circles and points

Chapter 17. Circles and lines

Chapter 18. Two or more circles

Chapter 19. Circles and angles

Chapter 20. A circle and a triangle

Chapter 21. Circles and polygons

Chapter 22. Circumference and arc

Chapter 23. Disks and sectors

Chapter 24. Overview of Chapter IV